Biomedical Image Processing

MEDICAL IMAGING MODALITIES

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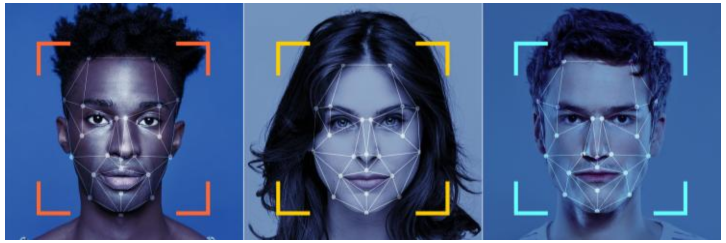
Medical Imaging

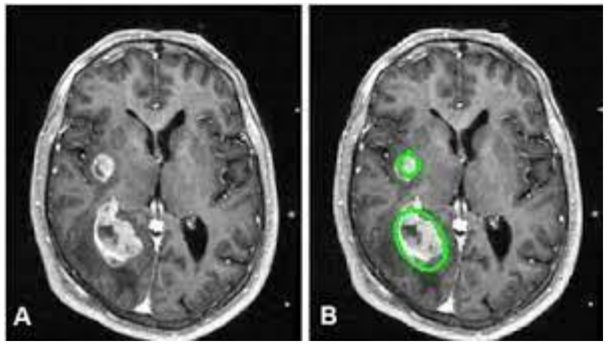
GENERAL IMAGE PROCESSING APPLICATIONS

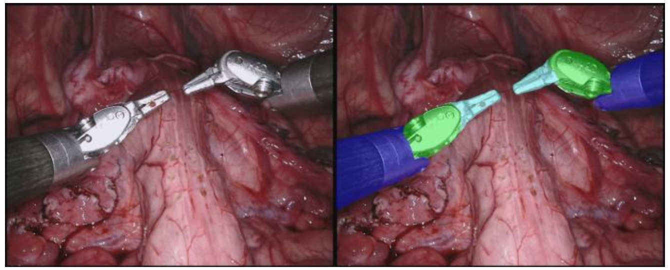
- Image processing can be used to:
  - Enhancement and restoration of images;
  - Image composition;
  - Detection of regions of interest (e.g. security – facial recognition);
  - Image encoding and data compression, ...



MEDICAL IMAGE PROCESSING APPLICATIONS

- Medical image processing can be used to:
  - Disease diagnosis;
  - Pathology detection;
  - Monitoring of the patient's condition;
  - Segmentation of anatomical structures;
  - Planning/guiding surgical interventions, ...





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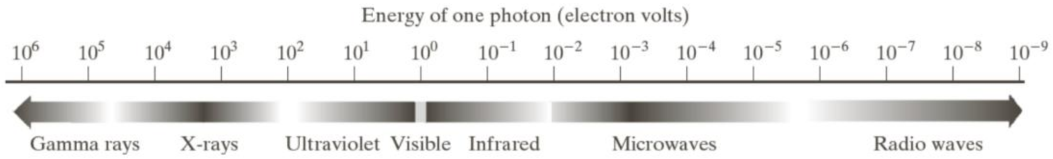
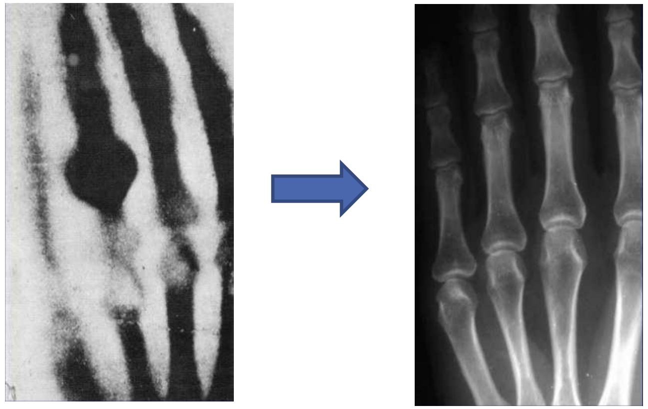
HISTORY

- Wilhelm Conrad Roentgen (German physicist) discovered the x-ray in 1895;



- This discovery made it possible to “see” the inside of the body due to the interaction of the x-ray with the tissues;
- Since then, the field of medical imaging has evolved, with new medical imaging modalities emerging and improving them;

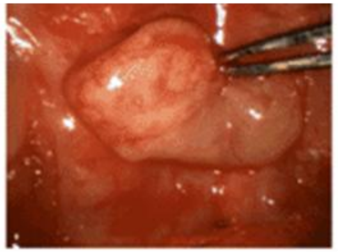


Medical Imaging

CAD- COMPUTER AIDED DIAGNOSIS

- Image processing applications are not autonomous in medicine;
- They are support instruments for health professionals;
- Medical images are auxiliary diagnostic tests;
- Even using automatic methods to evaluate the image, the final confirmation is always given by the doctor::
  - This is valid for the diagnosis of pathologies;
  - But it is also valid for surgical interventions in real time;

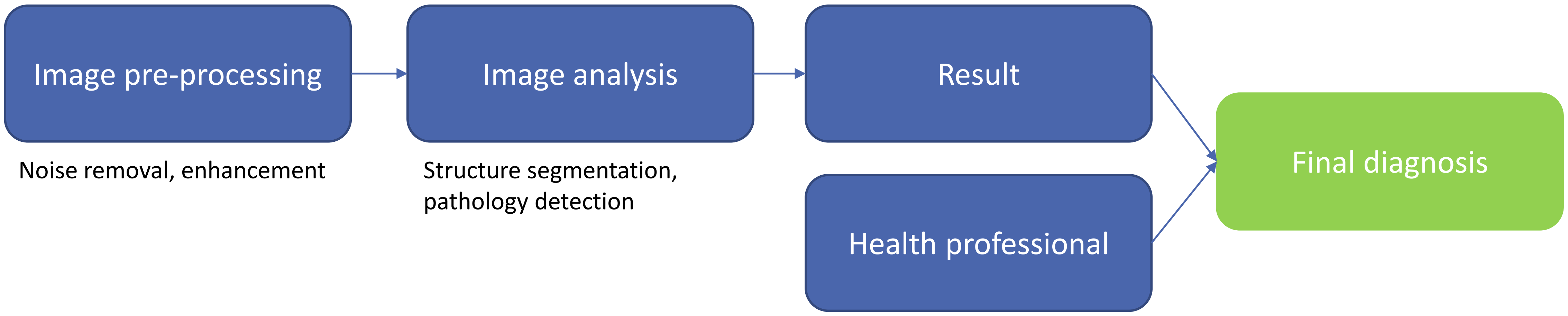





Medical Imaging

CAD- COMPUTER AIDED DIAGNOSIS

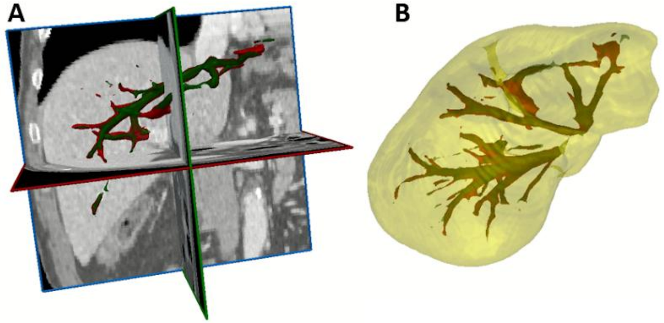
- Development of programs to be used by health professionals;
- These types of programs contain image processing methods to recognize abnormal patterns and warn healthcare professionals;
- They help to improve pathology detection: reduction of false negatives!
- In the case of quantification of volumes/pathologies, they avoid completely manual delineation, which is very expensive.



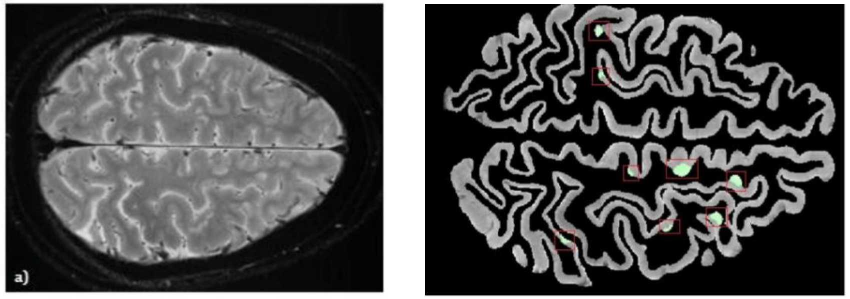
Medical Imaging

CAD- COMPUTER AIDED DIAGNOSIS – EXAMPLES

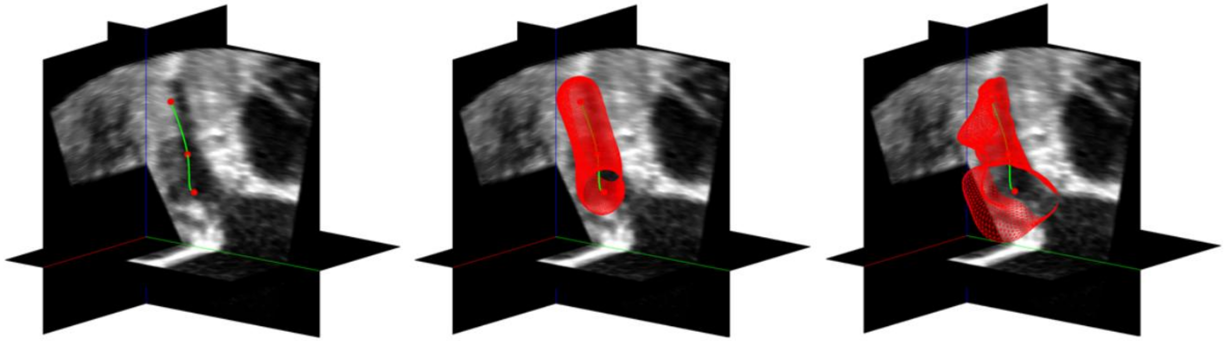
Detection of calcifications on mammograms



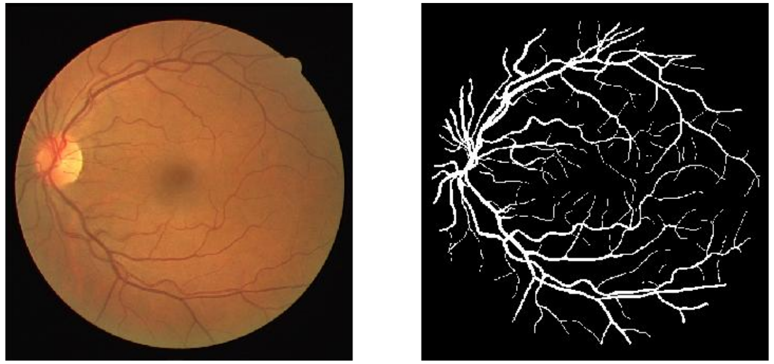
Segmentation of the liver and hepatic vessels in Computed Tomography images



Detection of multiple sclerosis on MRI images



Semiautomatic segmentation of the ventricular appendage

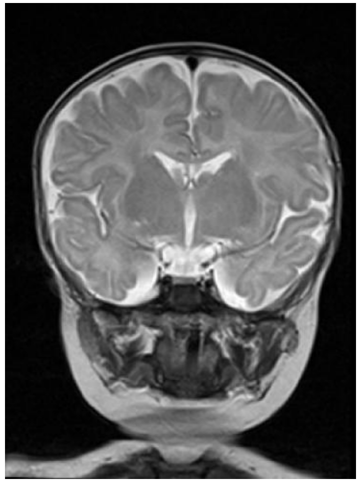


Delineation of retinal vessels

Types of Imaging Modalities

IMAGING MODALITIES

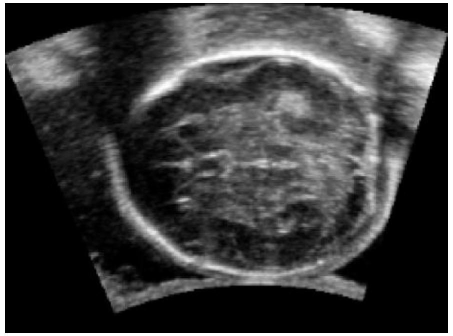
- Ionizing radiology
  - X-Ray;
  - CT – Computed Tomography;
  - Nuclear medicine;
- MRI – Magnetic Ressonance Imaging;
- US - Ultrasound
- Video
  - Probes with cameras (endoscopy, colonoscopy, laparoscopy, ...)



MRI

CT

X-Ray



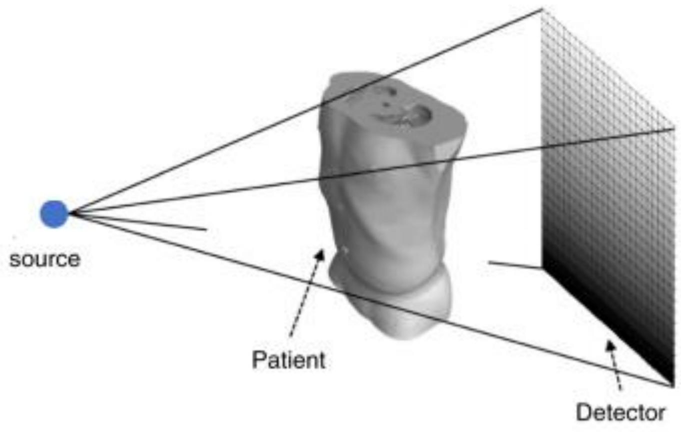
US

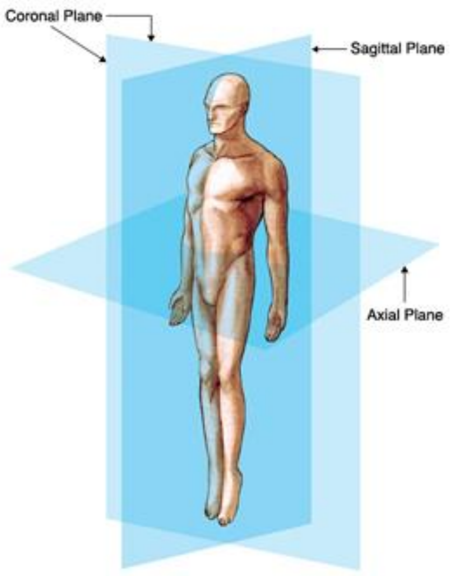
EXAMPLE OF MODALITIES VARIANTS

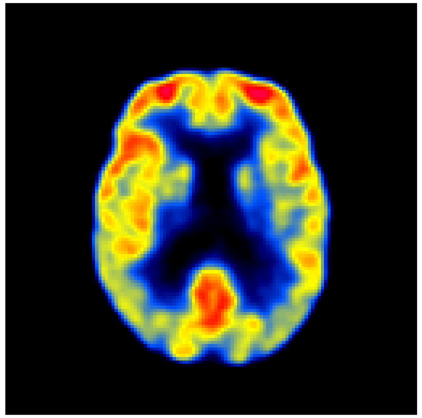
- Radiology - mammography, angiography, fluoroscopy, ...
- US - ecodoppler

Types of Imaging Modalities

IMAGING MODALITIES







Projection-based

X-ray;
Mammography;
Angiograph;
Fluoroscopy.

Cross-sectional

CT;

MRI;

US.

Funcional

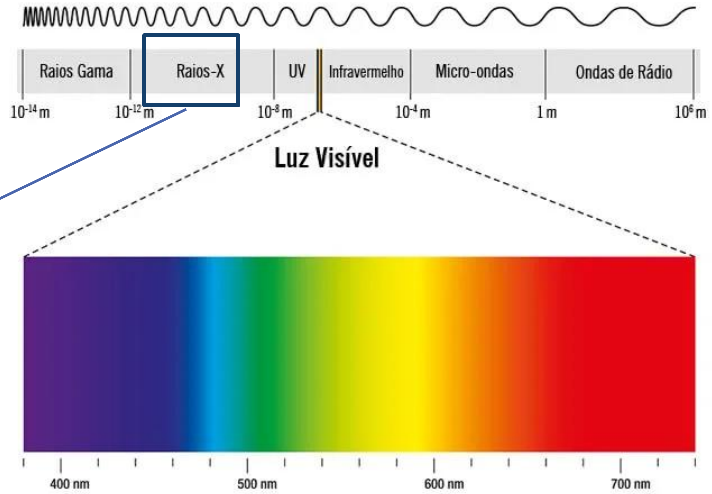
MRI funcional;

Doppler.

Types of Imaging Modalities – X-Ray

RADIOGRAPHY - CONCEPTS

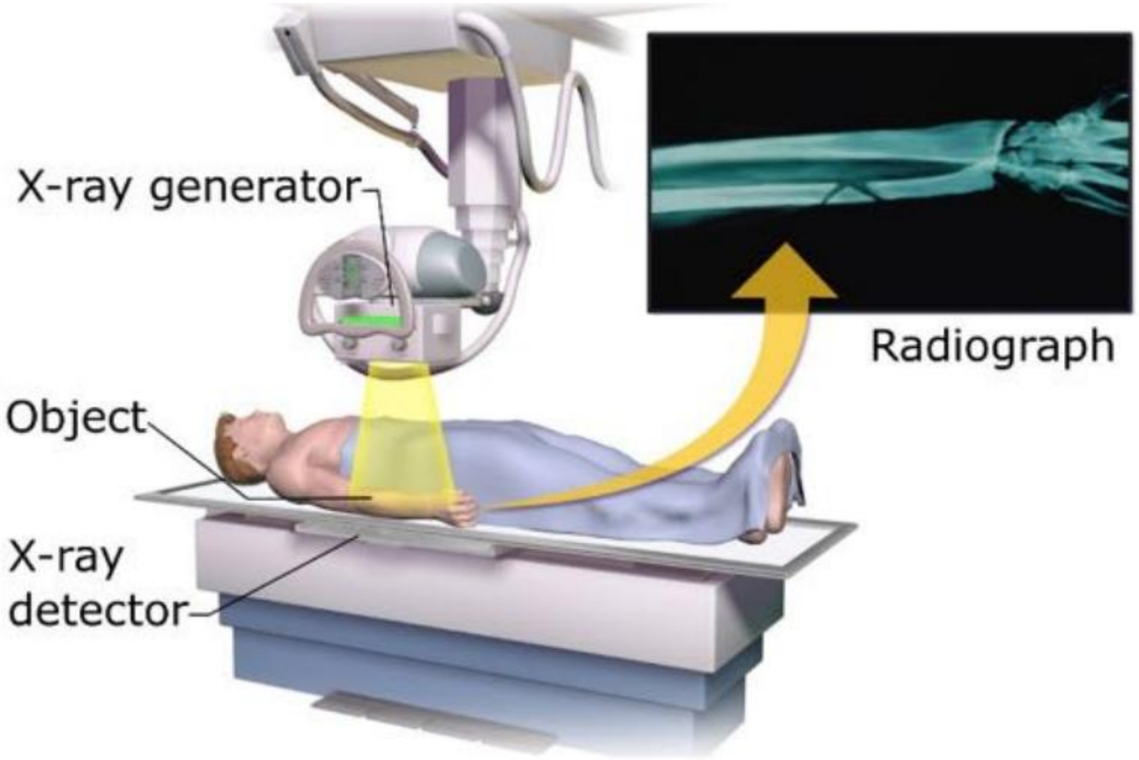
- Pioneering system used to obtain images of the inside of the human body;
- Radiography was one of the first uses of X-radiation since its discovery by Roentgen in 1895;
- Based on electromagnetic radiation X-rays of the electromagnetic spectrum;
- Image generation is obtained due to the interaction of X-radiation with human tissues.



Due to their short wavelength, X-rays are able to penetrate structures and materials that are opaque to visible light.

Types of Imaging Modalities – X-Ray

RADIOGRAPHY – IMAGE ACQUISITION AND GENERATION



Types of Imaging Modalities – X-Ray

RADIOGRAPHY – IMAGE ACQUISITION AND GENERATION

1. A homogeneous X-ray beam is emitted towards the body;;
2. X-rays will interact with body tissues, being absorbed/attenuated by tissues:
    1. Tissues with lower densities (air, fat) have a lower absorption;
    2. Tissues with greater intensity (soft tissues, bones, metals) have a greater absorption.
3. The X-rays that are not absorbed by the tissues will be captured on a photographic film, forming an image;
4. Photographic films can also be replaced by photosensitive detectors to form a digital image.

Types of Imaging Modalities – X-Ray

RADIOGRAPHY – CONTRAST

- Many organs have similar permeabilities to X-rays and are therefore indistinguishable on a radiograph;
- Radiology can use substances that have different permeabilities to X-rays than different parts of the body;
- If an element with a different permeability is introduced into a certain organ or cavity, that organ or cavity will become visible;
- These are called contrast agents:
  - Barium
  - Iodine

Types of Imaging Modalities – X-Ray

X-RAY - VARIANTS

MAMMOGRAPHY

- Specific for breast examination;
- Low energy X-ray.

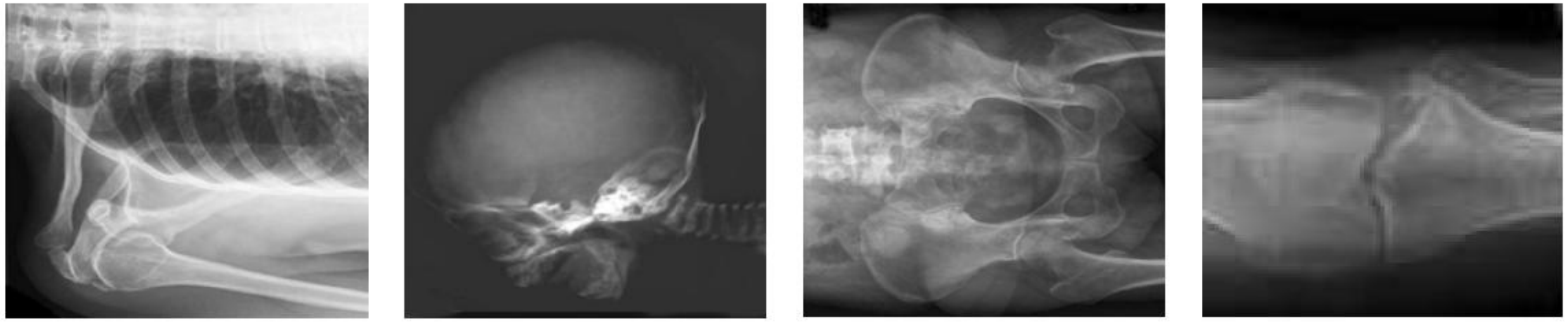
FLUOROSCOPY

- Visualization in real time, with the images being generated continuously and visualized;
- The X-ray emission is continuous;
- Digital imaging;

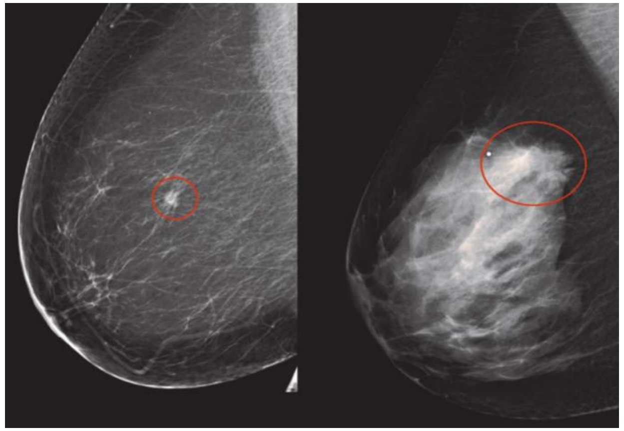
ANGIOGRAPHY

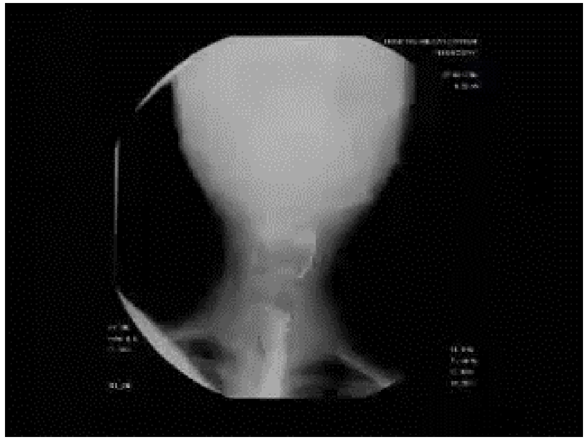
- Use of contrast to visualize the circulation of blood vessels;

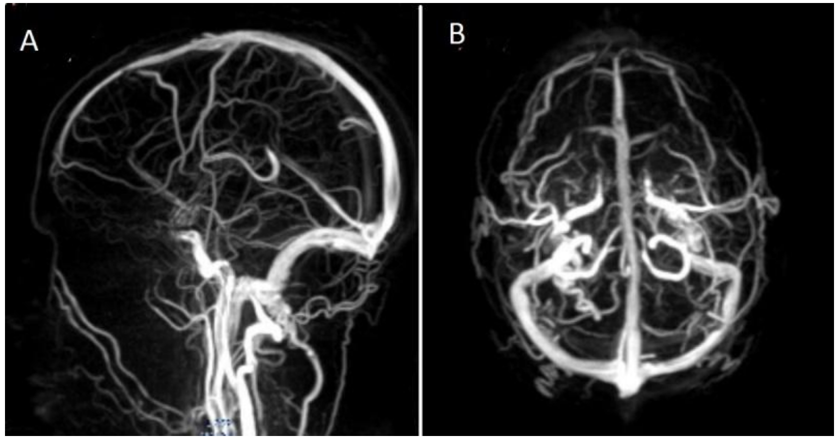
Types of Imaging Modalities – X-Ray



Shoulder Head Pelvis Knee







Types of Imaging Modalities – X-Ray

RADIOGRAPHY – ADVANTAGES/DISADVANTAGES

ADVANTAGES

- Non-invasive, fast, painless;
- Can be used for surgical planning:
  - E.g. Can be used when inserting catheters into the body.

DISADVANTAGES

- Exposes the patient to radiation:
  - May cause cancer (high doses);
  - Tissue damage (high doses);

Types of Imaging Modalities – X-Ray

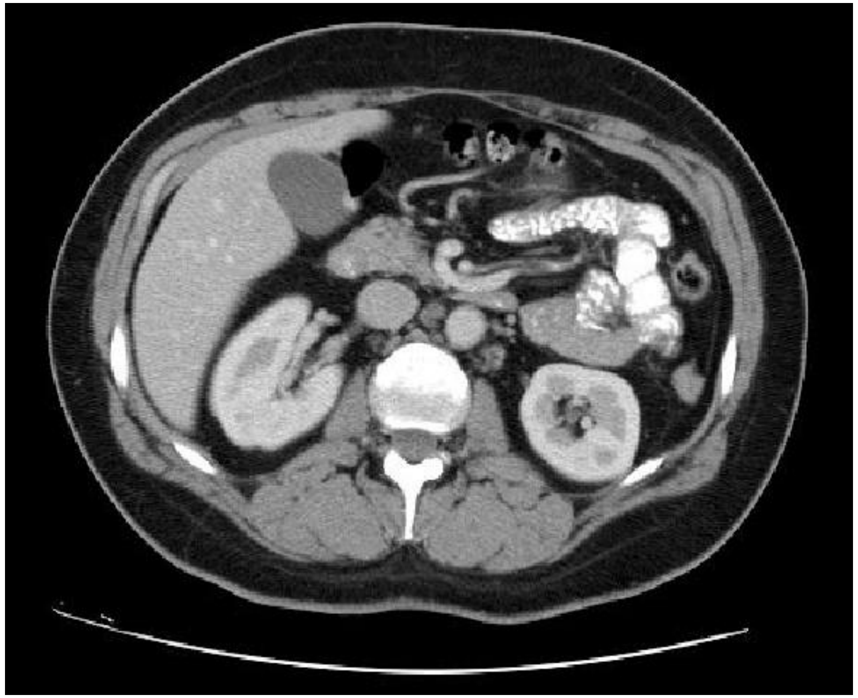
RADIOGRAPHY – MEDICAL APPLICATIONS

- Musculoskeletal and dental examinations;
- Determine the type and extent of bone fractures;
- Detect pathologies in the lungs;
- Visualization of the stomach and intestines;
- Fluoroscopy allows checking the movement of organs (stomach, intestine and colon);
- Study the cardiac and cerebral vessels;
- Mammography is used to diagnose breast lesions;

Types of Imaging Modalities – CT

COMPUTED TOMOGRAPHY - CONCEPTS

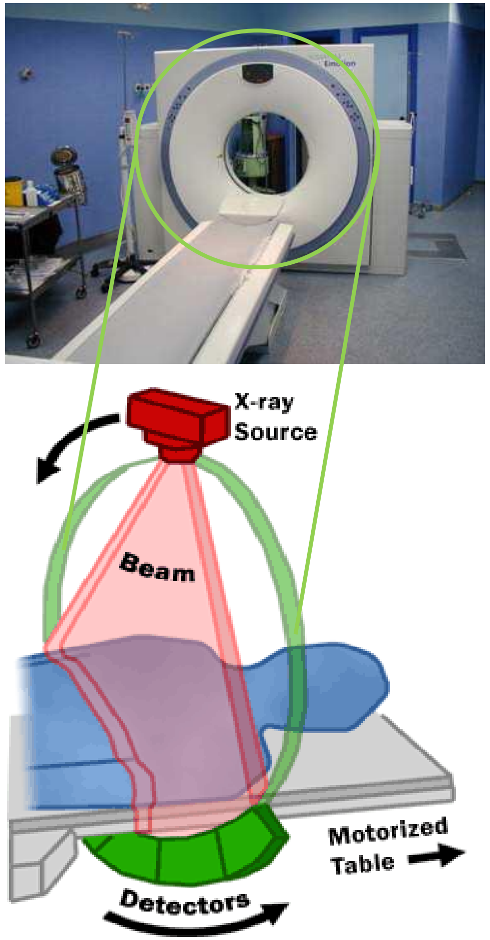
- Early 70's;
- As in radiology, it uses X-ray radiation;
- Basic idea - using several radiographs from multiple perspectives to build images based on these perspectives;
- However, it is a cross-sectional image, not a projection;
- First modality entirely based on digital image reconstruction;



Types of Imaging Modalities – CT

COMPUTED TOMOGRAPHY – IMAGE ACQUISITION AND GENERATION

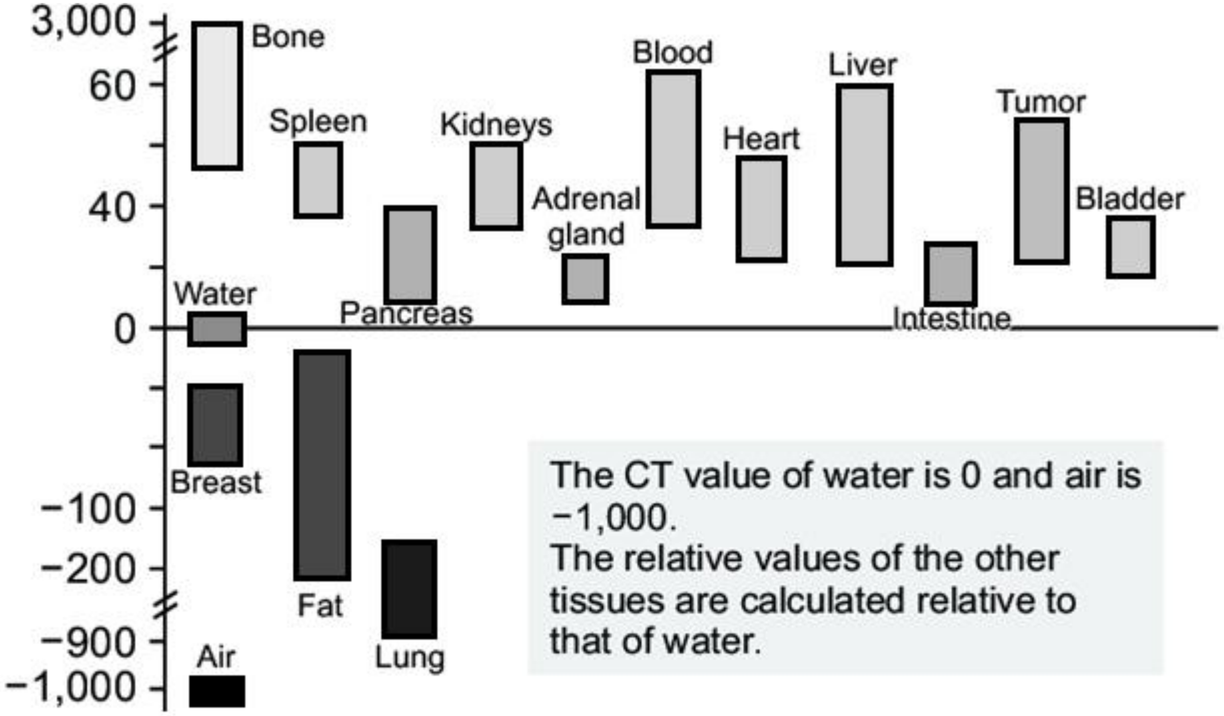
- A CT scan contains a circular ring where the patient is inserted.
- Contains a ring composed of an x-ray emitter and a detector on opposite sides, which rotate simultaneously;
- During the examination, a X-ray beam is emitted which is detected on the detector after passing through the tissues;
- This happens for various body positions;
- This process allows generating multiple views of the body, forming 3D images..



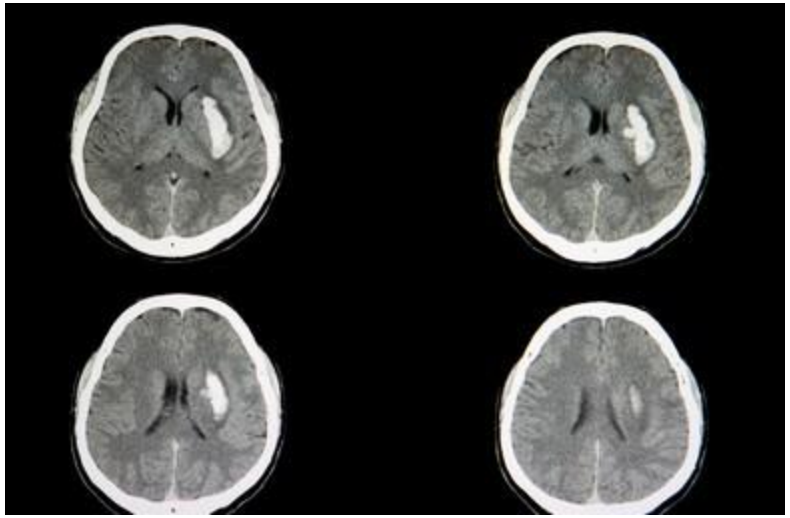
Types of Imaging Modalities – CT

COMPUTED TOMOGRAPHY - IMAGES

- The images generated in CT have a fixed scale of intensities (value of each pixel) and are given in Hounsfield units.
- The reference material for CT values is water (value=0)
  - Tissues with attenuation greater than water have positive values
  - Tissues with less attenuation than water have negative values



Types of Imaging Modalities – CT



Types of Imaging Modalities – CT

COMPUTED TOMOGRAPHY - ADVANTAGES/DISADVANTAGES

ADVANTAGES

- Non-invasive, fast, painless;
- Good spatial resolution;
- Can distinguish small differences in densities;

DISADVANTAGES

- High exposure to radiation;
- It's not real-time;
- Contrast usually has to be used;

Types of Imaging Modalities – CT

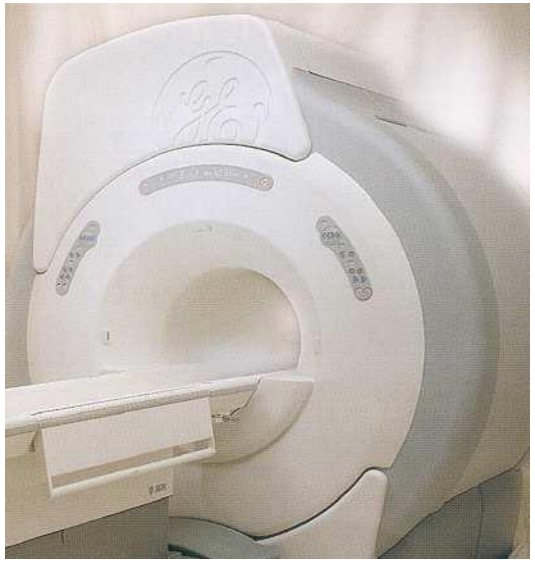
COMPUTED TOMOGRAPHY - MEDICAL APPLICATIONS

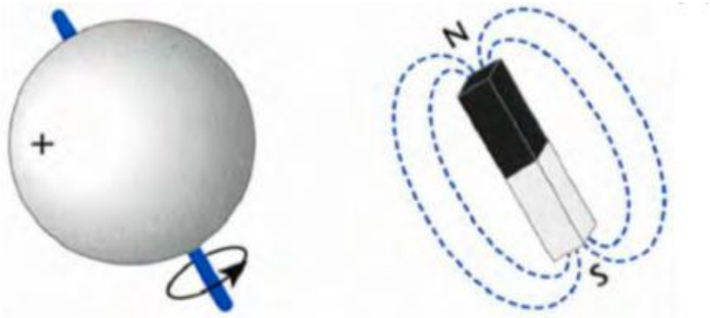
- Used for many parts of the body: bones, abdominal, dental, cervical, ...;
- Diagnosis of abnormalities;
- Surgical planning;
- Cancer treatment monitoring.

Types of Imaging Modalities– MRI

MAGNETIC RESONANCE IMAGING - CONCEPTS

- MRI uses the concepts of magnetic fields and radiofrequency to acquired images inside the body;
- To do so, it is based on the matter's ability to absorb and emit RF energy under certain circumstances;

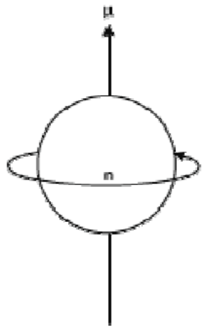


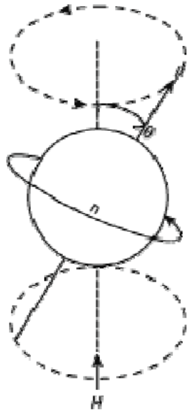


Types of Imaging Modalities– MRI

MAGNETIC RESONANCE IMAGING – IMAGE ACQUISITION AND GENERATION

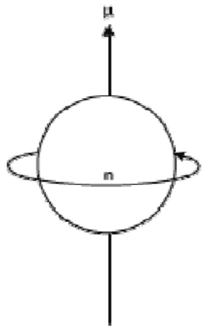
- SPIN: property of atoms consisting in the production of an angular momentum;
- The human body is composed of 63% hydrogen;
- When hydrogen is exposed to a magnetic field, it tends to line up with this field;
- In MRI, the patient is subjected to a permanent magnetic field that makes their hydrogen atoms line up;
- This field can be disturbed by radiofrequency pulses that cause atoms to spin as they absorb energy;
- When the pulses end, the hydrogen atoms return to their normal state (aligned with the field), releasing energy (relaxation);



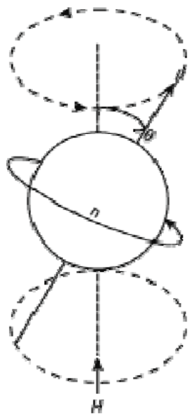


Types of Imaging Modalities– MRI

MAGNETIC RESONANCE IMAGING – IMAGE ACQUISITION AND GENERATION



- The relaxation time (and amount of energy) can be measured to produce the image:
  - Different tissues produce different intensity values according to the amount of hydrogen atoms they contain;
  - Different relaxation times can also be measured to generate different images within the resonance modality:
    - T1 and T2.



Types of Imaging Modalities– MRI

T2

T1

Types of Imaging Modalities– MRI

MAGNETIC RESONANCE IMAGING - ADVANTAGES/DISADVANTAGES

ADVANTAGES

- Non-invasive and painless;
- Does not use radiation;
- High spatial resolution;
- Great contrast of different tissues;

DISADVANTAGES

- Time consuming;
- Expensive;
- High magnetic field (excludes pacemakers and metallic prostheses);
- The images do not have a fixed intensity (between different devices for example).

Types of Imaging Modalities– MRI

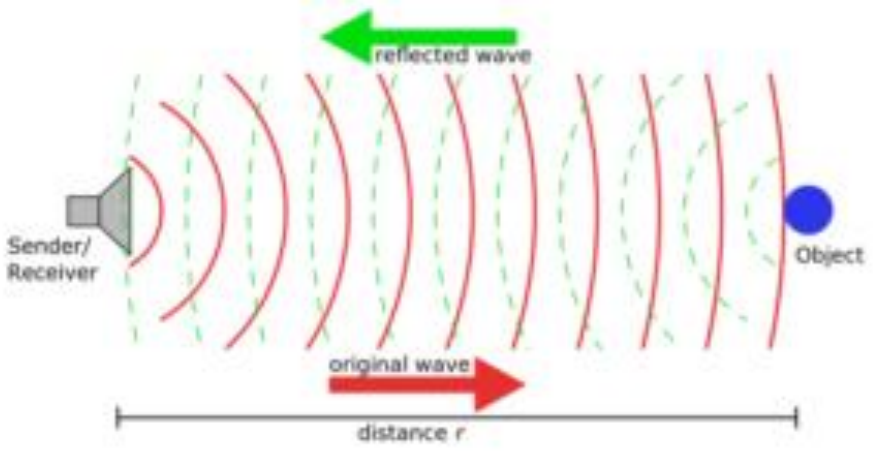
MAGNETIC RESONANCE IMAGING – MEDICAL APPLICATIONS

- Diagnosis of abnormalities in the brain and spine;
- Detection of tumors in various regions of the body;
- Detection of abnormalities in tendons and articulations;
- Surgical planning;

Types of Imaging Modalities– US

ULTRASOUND - CONCEPTS

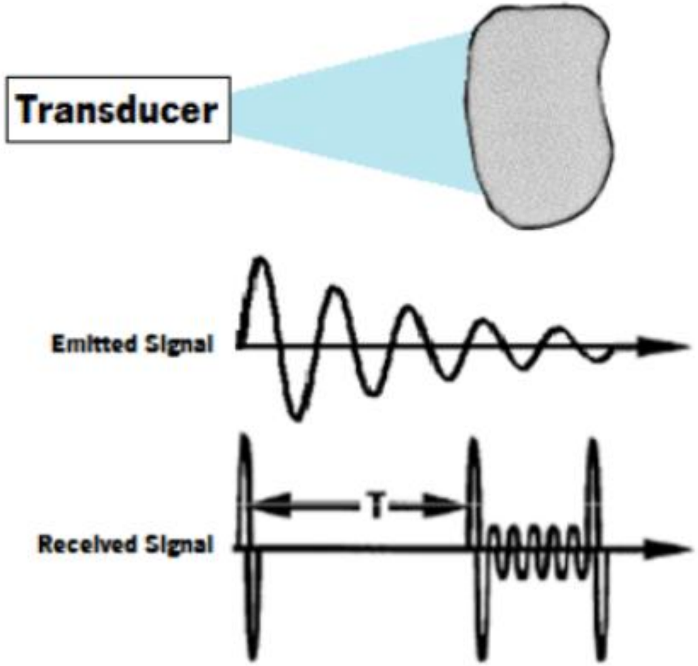
- Diagnostic technology that uses high-frequency waves reflected from tissues;
- Ultrasound is based on the behavior of sound and its interaction with objects;
- The boundary between media with different acoustic impedance induce reflection and refraction effects;
- Medical image acquisition using ultrasound uses the echoes resulting from reflections.



Types of Imaging Modalities– US

ULTRASOUND – IMAGE ACQUISITION AND GENERATION

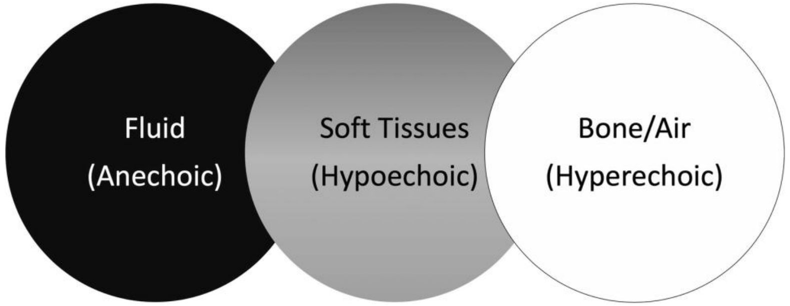
- A transducer (probe) emits a high-frequency wave;
- The waves are reflected by the tissues. More precisely, waves are reflected when they encounter a change in medium;
- The greater the impedance difference between the tissues, the greater the reflection;
- The transducer receives the reflections and the time that the signal takes to return is used to form the image..

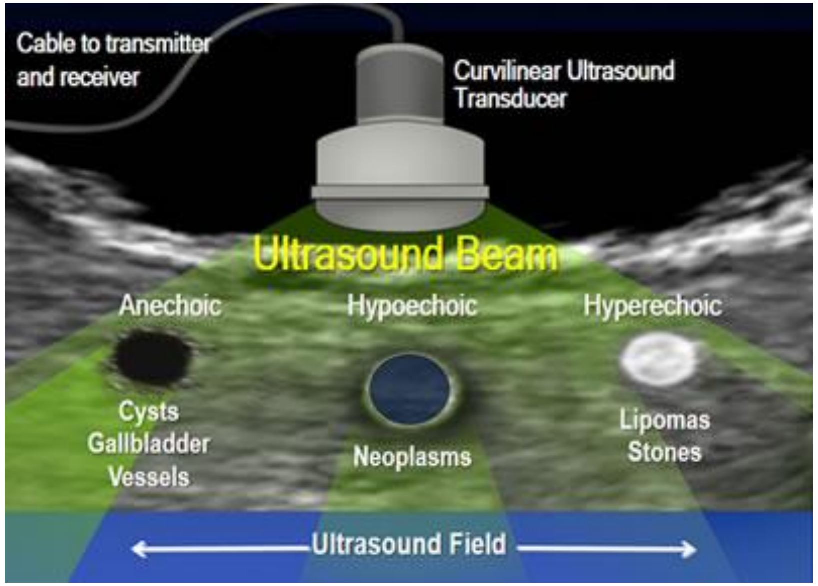


Types of Imaging Modalities– US

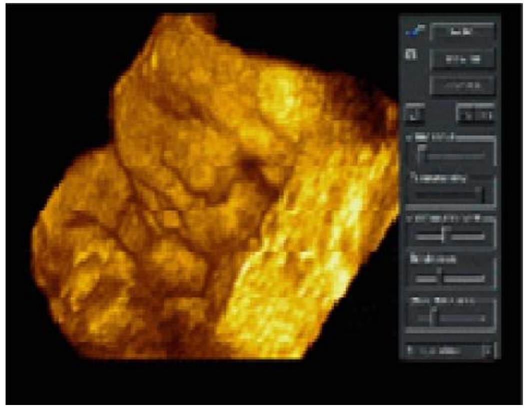
ULTRASOUND – IMAGES

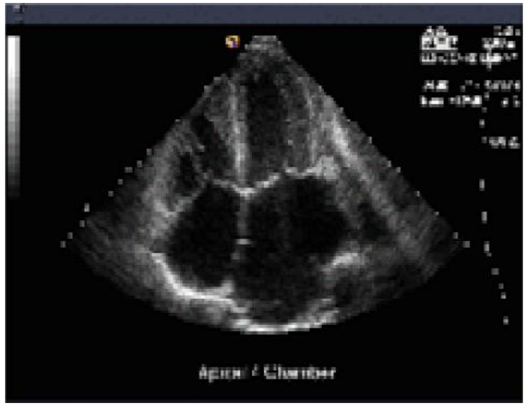
- Echogenicity – ability of tissues to reflect or transmit ultrasound waves (relative to surrounding tissues)

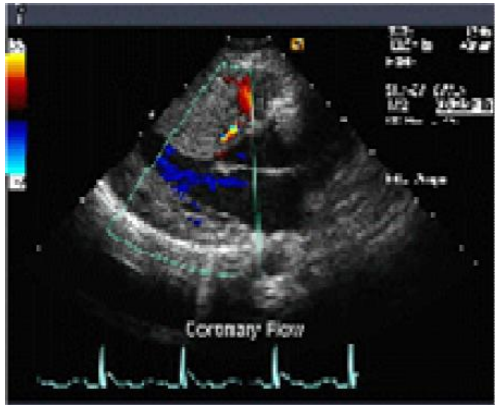




Tipos de Modalidades Imagiológicas – US

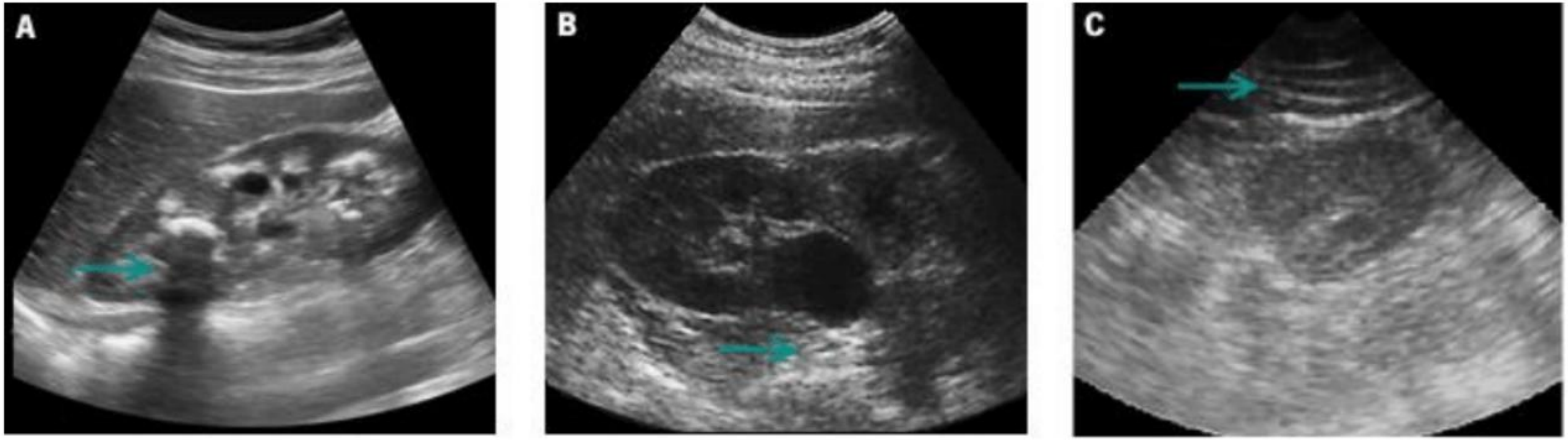






Tipos de Modalidades Imagiológicas – US

ULTRASOUND – ARTIFACTS



mages subject to a lot of artifacts: shadows, acoustic reinforcement; reverberation; …

Speckle Noise: Grainy image texture!

Tipos de Modalidades Imagiológicas – US

ULTRASOUND – ADVANTAGES/DISADVANTAGES

ADVANTAGES

- Non-invasive and no side effects;
- Low cost;
- Acquisition in real time;
- Great contrast of different tissues;

DISADVANTAGES

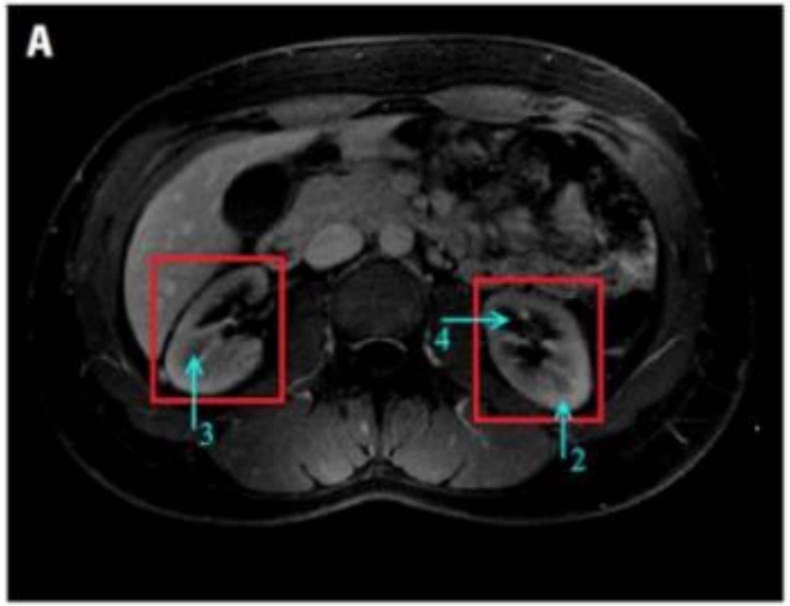
- Difficult acquisition;
- Very dependent on user experience;
- Very challenging image to process;

Types of Imaging Modalities– US

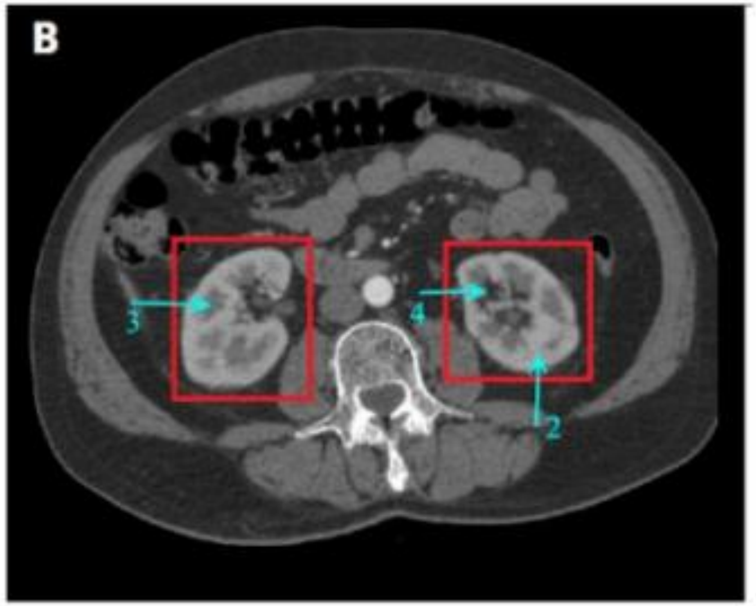
ULTRASOUND – MEDICAL APPLICATIONS

- First line of diagnosis;
- Analysis of solid organs, such as pancreas, aorta, kidney, liver, gallbladder,...
- Fetal development during pregnancy;
- Cardiac ultrasound;
- Guidance during interventions (insertion of needles);

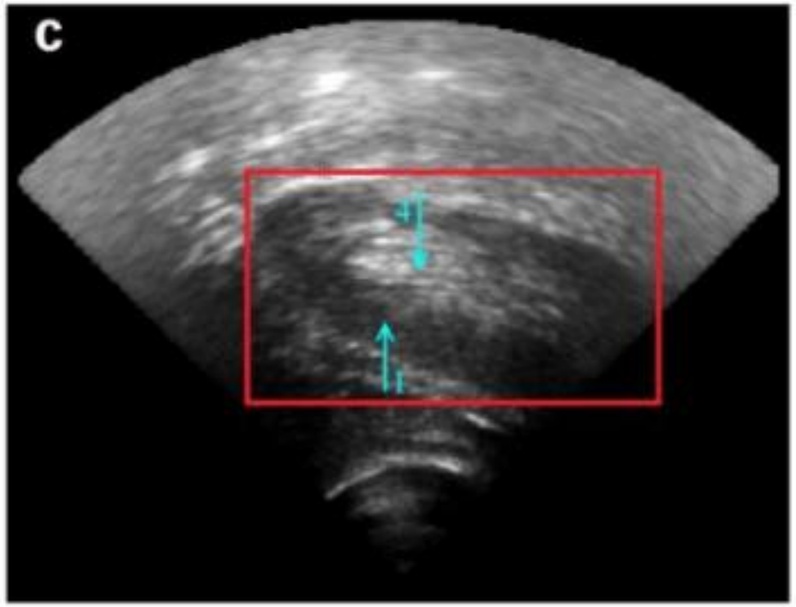
Types of Imaging Modalities



MRI



CT



US